MILITARY SPECIFICATION

SULFANILAMIDE, ALLANTOIN, AND AMINACRINE HYDROCHLORIDE SUPPOSITORIES, Vaginal

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers Sulfanilamide, Allantoin, and Aminacrine Hydrochloride Vaginal Suppositories (see 6.2).

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents, of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

PPP-C-186 - Containers, Packaging and Packing for Drugs, Chemicals and Pharmaceuticals.

STANDARDS

FEDERAL

FED-STD-285 - Capsules (for Medicinal Purposes).

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of specifications and standards required by contractors in conneciton with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Headquarters, Defense Personnel Support Center, ATTN: Directorate of Medical Materiel, DPSC-ATT, 2800 South 20th Street, Philadelphia, PA 19101, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 6505

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2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposals shall apply.

U. S. DEPARTMENT OF HEALTH, EDUCATION, AND WELFARE,
FOOD AND DRUG ADMINISTRATION (FDA)

Federal Food, Drug, and Cosmetic Act and Regulations Promulgated Thereunder.

(Application for copies should be addressed to the Food and Drug Administration, Washington, DC 20204.)

U. S. PHARMACOPEIAL CONVENTION, INC.

The United States Pharmacopeia (USP).

(Application for copies should be addressed to the Mack Publishing Company, Easton, PA 18042.)

The National Formulary (NF).

(Application for copies should be addressed to the Mack Publishing Company, Easton, PA 18042.)

3. REQUIREMENTS

3.1 Material. Shall be in accordance with all applicable requirements of FED-STD-285, and as specified herein.

S2. Classification. Shall be type II.

S5.4 The following is added to this paragraph:

Shall be a vaginal suppository containing 1.05 gram sulfanilamide, 0.14 gram allantoin, and 0.014 gram of aminacrine hydrochloride, and other suitable ingredients, within the specified limits. Ingredients and preservative(s) shall be acceptable to the FDA in the concentrations used.

S6.2.14 Water resistance. Delete this paragraph in its entirety. There is no substitute.

S6.2.15 Acid solubility. In the last paragraph delete "5 minutes" and substitute "15 minutes".

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S6.4.2 Inprocess testing. Delete this paragraph in its entirety and substitute.

"For soft gelatin suppositories, not less than 10 capsules shall be drawn from each lot at the beginning of the filling operation and tested for fill weight. If filling pumps are adjusted or changed, the fill weight test shall be repeated. In addition, for soft gelatin suppositories, the seal thickness shall be verified at the beginning and during the encapsulating (filling) operation. The inprocess testing shall be applicable to suppositories from each encapsulating machine. The test results shall be recorded for each encapsulating machine used for each lot to confirm uniformity of weight."

3.2 Assay of the finished suppository.

3.2.1 Sulfanilamide. Shall contain not less than 90.0 percent and not more than 110.0 percent of the required amount, when determined as specified in 4.5.1.

3.2.2 Allantoin. Shall contain not less than 90.0 percent and not more than 110.0 percent of the required amount, when determined using a suitable, accurate, and reproducible method.

3.2.3 Aminacrine hydrochloride. Shall contain not less than 90.0 percent and not more than 110.0 percent of the required amount, when determined using a suitable, accurate, and reproducible method.

3.3 Identification of the active ingredients in the finished suppository.

3.3.1 Sulfanilamide. An orange precipitate shall result in accordance with the procedure specified in 4.5.2.

3.3.2 Allantoin and aminacrine hydrochloride. Assurance of the presence of allantoin and aminacrine hydrochloride in the finished suppositories shall be determined by suitable procedures.

3.4 pH. The pH shall be not less than 4.0 and not more than 5.5 when contents of 2 suppositories are added to 15 ml of purified water and determined potentiometrically at 25° C, using the USP method.

3.5 Ingredients.

3.5.1 Sulfanilamide. Shall be in accordance with the tests, standards, and requirements of the NF XI, which was official from 1 October 1960 until 31 August 1965, except the melting range shall be 163.5° C - 166.5° C.

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3.5.2 Allantoin. Shall comply with the following:

3.5.2.1 Loss on drying. Shall lose no more than 0.2 percent when approximately 2 grams are accurately weighed and dried overnight over silica gel, under vacuum.

3.5.2.2 Residue on ignition. Shall not exceed 0.05 percent when approximately 2 grams are accurately weighed and ignited to constant weight in a tared crucible.

3.5.2.3 Substances darkened by sulfuric acid. Solution shall remain colorless when 0.5 gram is dissolved in 5 ml of sulfuric acid TS, USP.

3.5.2.4 Chloride. Shall not exceed 0.03 mg chloride (0.006 percent) when determined as follows:

Dissolve 1.5 grams in 75 ml of boiling purified water, cool to room temperature, and restore to a volume of 75 ml with purified water. Mix well and filter. To 25 ml of filtrate, add 0.5 ml of nitric acid and 1 ml of silver nitrate, TS. When compared in matched Nessler tubes, the opalescence of the sample shall not exceed that of a standard containing 0.03 mg of chloride in 25 ml purified water to which has been added 0.5 ml nitric acid and 1 ml of silver nitrate.

3.5.2.5 Sulfate. Shall not exceed 0.1 mg sulfate (0.02 percent) when determined as follows:

Evaporate 25 ml of the filtrate obtained in the chloride test to less than 10 ml. Cool to room temperature, filter, if necessary, and dilute to 10 ml with purified water. Add 0.5 ml of 1N hydrochloric acid and 1 ml of barium chloride TS. When compared in matched Nessler tubes, the turbidity of the sample shall not exceed that of a standard containing 0.1 mg of sulfate in approximately 10 ml of purified water to which has been added 0.5 ml of 1N hydrochloric acid and 1 ml of barium chloride TS.

3.5.2.6 Assay. Shall contain not less than 98.5 percent and not more than 102.0 percent calculated on a dried basis, when determined as follows:

Accurately weigh approximately 250 mg of sample previously dried over silica gel using vacuum, into a glass-stoppered flask. Pipet 50 ml of anhydrous ethylenediamine into the flask, stopper, and shake until the sample is dissolved. Titrate with a solution of 0.1N sodium methoxide in benzene, volumetric solution, using 1 ml of 0.1 percent o-nitroaniline (in benzene) indicator, until the solution changes from yellow to orange. Perform a blank determination and make any necessary corrections. Each ml of 0.1N sodium methoxide is equivalent to 15.81 mg of allantoin. Calculate the percent allantoin in the sample.

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3.5.3 Aminacrine hydrochloride. Shall comply with the following:

3.5.3.1 Mesh size. Not less than 100 percent shall pass through a U.S. Series No. 80 sieve, when determined using the USP method.

3.5.3.2 Loss on drying. Shall not exceed 8.0 percent when approximately 1 gram of sample is accurately weighed and dried to constant weight at 110° C.

3.5.3.3 Arsenic. Shall not exceed 10 parts per million when determined using the NF method.

3.5.3.4 Assay (aminacrine base). Shall contain not less than 82.5 percent and not more than 84.5 percent calculated on a dried basis, when determined as follows:

Solutions. Prepare all solutions using purified water.

Accurately weigh approximately 500 mg of sample into a 100 ml volumetric flask. Add 10 ml of 20 percent reagent sodium acetate solution. Dilute to about 50 ml, stopper flask and shake until sample is dissolved. Add an excess of 10 percent sodium hydroxide solution to free the base. Swirl for several minutes and permit to stand for 15 minutes, occasionally agitating the solution. Dilute to volume with purified water. Mix well and filter the mixture through a previously prepared tared filter crucible, catching the filtrate in a clean dry vacuum flask. Transfer the undiluted filtrate to another flask and set aside to be used in the determination of hydrochloride. Rinse the 100 ml volumetric flask with purified water and transfer the rinsings to the filter crucible. Wash the precipitated base with several small portions of purified water until the washings are no longer alkaline to phenolphthalein TS. Dry to constant weight and calculate the precentage of base obtained calculated on the dry basis. Save the sample of base for melting point determination.

3.5.3.5 Melting point of base. Shall melt between 233° - 237° C, when determined using the USP Method Ia, with the sample inserted at approximately 228° C, and the temperature of the bath increased at the rate of 1° C per minute.

3.5.3.6 Hydrochloride. Shall contain not less than 15.25 percent and not more than 16.00 percent, calculated on a dried basis, when determined as follows:

Transfer exactly 75 ml of the undiluted filtrate from the assay to clean, dry 250 ml iodine flask. Render the solution acid to litmus with dilute nitric acid. Add 10 additional ml of the acid, plus exactly 50.0 ml of 0.1N Silver Nitrate Volumetric Solution. Add 3 ml of nitrobenzene and shake well. Rinse the stopper and walls of the flask with purified water. Add approximately 3 ml of ferric ammonium sulfate TS, and mix well. Titrate the excess

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silver nitrate with 0.1N Ammonium or Potassium Thiocyanate Volumetric Solution until the characteristic flesh color appears. Each ml of 0.1N silver nitrate is equivalent to 3.646 mg of hydrochloride.

Calculation:

$$(A - B) \times 3.646 \times \frac{100}{75} \times \frac{100}{\text{Sample (mg)}} \times \frac{100}{(100 - \text{percent LOD})} = \text{percent hydrochloride}$$

where:

A = 50.0 ml of 0.1N $AgNO_3$
B = ml of 0.1N NH_4SCN or KSCN consumed.

(Make appropriate corrections for exact normalities of volumetric solutions.)

3.6 Other ingredients. All other ingredients entering into the suppositories shall comply with S5.2 of FED-STD-285.

3.7 Expiration dating period. The expiration dating period shall be not less than 36 months.

4. QUALITY ASSURANCE PROVISIONS

4.1 Supplier responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and, (i) as to any expiration dated item, for the expiration dating period specified by the contractor for such item, or such longer period as may be required by regulation of any federal agency; or (ii) as to non-dated items, for not less than 3 years after delivery of the items to the Government.

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4.1.2 No company supplying any ingredient(s) to the contractor will be considered an acceptable facility for the performance of any inspection requirements specified herein.

4.2 Packaging inspection. The inspection of the intermediate package, packing and marking for shipment and storage shall be in accordance with quality assurance provisions of the applicable container specification and the marking requirements of MIL-STD-129.

4.3 Labeling inspection. Sampling shall be in accordance with MIL-STD-105 and the unit of product shall be the unit of issue. The inspection lot shall be the number of units of issue of each batch, lot, control number, etc., offered for inspection at one time. The inspection level for labeling shall be S-2 and the acceptance number shall be zero.

4.4 Sampling. The sampling as shown in FED-STD-285 shall be applicable.

4.5 Tests.

4.5.1 Assay of sulfanilamide in the finished suppository. To a 1 liter volumetric flask, add 400 ml of purified water and 100 ml of 1N hydrochloric acid, and mix. Place the flask on a steam bath and warm the solution to approximately 60° C. Add five suppositories to the solution and swirl. Continue heating until the suppositories are completely in solution. Cool the solution to 25° C and dilute to volume with purified water. Accurately pipet a 100.0 ml aliquot of the solution into a 400 ml beaker. Add 10 ml of concentrated hydrochloric acid and cool to 15° C. Add about 25 grams of crushed ice, and slowly titrate with 0.1M Sodium Nitrite Volumetric Solution, stirring vigorously, until a blue color is produced immediately when a glass rod dipped into the titrated solution is touched on starch iodine paper. When the titration is complete, the end-point is reproducible after the mixture has been allowed to stand for 1 minute. Each ml of 0.1M sodium nitrite is equivalent to 0.01722 gram of sulfanilamide. Calculate mg of sulfanilamide in the suppository.

4.5.2 Identification of sulfanilamide in the finished suppository.

Reagents.

Prepare all reagent solutions in purified water unless otherwise indicated.

4 percent sodium hydroxide.
10 percent hydrochloric acid.
1 percent sodium nitrite.
50 mg of betanaphthol in 2 ml of 10 percent sodium hydroxide solution.

Procedure.

Place the content of one suppository in a 150 ml beaker, add 50 ml of 4 percent sodium hydroxide solution and swirl to obtain solution. Place 1 ml of this sample preparation in a test tube immersed in an ice bath. Add 1 ml of 10 percent hydrochloric acid and mix. Add 4 ml of sodium nitrite solution and mix well. Then add one or two drops of betanaphthol solution.

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5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with all applicable requirements of PPP-C-186, and as specified herein.

5.2 Packaging. Packaging shall be level A or C, as specified (see 6.1).

5.2.1 Unit of issue. One package containing sixteen individually packaged suppositories along with an applicator, as specified, constitutes one unit of issue.

5.2.2 Level A.

5.2.2.1 Immediate container. The immediate container shall be commercially available immediate container considered acceptable by the Food and Drug Administration (FDA) for the product contained therein and shall be of such compatibility that neither the contents nor the container are altered in any way by each other. Each suppository shall be individually sealed.

5.2.2.2 Unit packaging. Each unit shall be packaged as specified in 5.2.5 of PPP-C-186. A suitable applicator for administering the suppositories shall be in each unit.

5.2.2.3 Intermediate packaging. Twelve (12) units of issue shall be packaged as specified in 5.2.7.1 of PPP-C-186..

5.2.3 Level C. Units shall be packaged in standard commercial containers of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose.

5.3 Packing. Packing shall be level A, B or C, as specified (see 6.1).

5.3.1 Level A. Ninety-six (96) units of issue shall be packed as specified in 5.4.4 of PPP-C-186.

5.3.2 Level B. Ninety-six (96) units of issue shall be packed as specified in 5.4.3.1.1 of PPP-C-186.

5.3.3 Level C. The subject commodity shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

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5.4 Marking.

5.4.1 Labeling. Labeling shall be in accordance with the requirements of the Federal Food, Drug and Cosmetic Act and shall include tne information required below:

5.4.1.1 Immediate container. Each immediate container label shall bear the label information in accordance with commercial practice.

5.4.1.2 Unit package. Each unit package label shall bear the following information:

(a) label information in accordance with commercial practice.

(b) the National Stock Number.

(c) the expiration date.

NOTES: (1) The National Stock Number shall not be preceded by any prefix.

(2) Strip labeling may be used.

(3) Label adhesion test shall not apply.

5.4.2 Intermediate package. Intermediate package shall be marked as specified in 5.5.3 of PPP-C-186. Type II shelf life markings as specified in MIL-STD-129 shall be shown. Marking shall include the legend: "STORE AT CONTROLLED ROOM TEMPERATURE 15^{0} - 30^{0} C (59^{0} - 86^{0} F)".

5.4.3 Exterior container. Exterior container shall be marked as specified in 5.5.4 of PPP-C-186. Type II shelf life marking as specified in MIL-STD-129 shall be shown. Marking shall include the legend: "STORE AT CONTROLLED ROOM TEMPERATURE 15^{0} - 30^{0} C (59^{0} -86^{0} F)".

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification.

(b) National Stock Number.

(c) Selection of applicable levels of packaging and packing (see 5.2 and 5.3).

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6.2 This specification covers the following item:

National Stock Number	Item Identification
6505-01-003-7695	SULFANILAMIDE, ALLANTOIN, AND AMINACRINE HYDROCHLORIDE SUPPOSITORIES, Vaginal, 16's

Custodians:
Army - MD
Navy - MS
Air Force - 03

Review activity:
DoD - MB

Preparing activity:
DLA - DM

Project No. 6505-1708

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

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DOCUMENT IDENTIFIER AND TITLE
MIL-S-37561 SULFANILAMIDE, ALLANTOIN, AND AMINACRINE HYDROCHLORIDE SUPPOSITORIES

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

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